

digital signal processing (DSP).

The SMT industry is growing steadily and is likely to see a surge in the demand for trained manpower.

Robotics is going to see wider adoption in industrial plants for automated material handling and production. Engineering students can avail this opportunity by considering a course in robotics.

Jayakrishnan T., director and country manager, Energid Robotics & Machine Vision, explains, "Obviously, the requirements in this field necessitate a quality learning centered around the fundamentals of physics, dynamics and mathematics in general. Specific to our company, we would like engineers to have some hardware integration experience and, of course, good software coding skills and vision algorithm knowledge too. Another thing is making the software easy to use from a non-coder standpoint. So the ability to create a good user interface (human factors) to simplify the process is always in demand."

"As far as freshers are concerned, the amount of hands-on experience with robotic hardware and integration would depend on the student's desire as well as the quality of the institution. Apart from the knowledge acquired from the institution, a shorter learning curve with good grasping, strong mathematics and ability to conceptualise a solution are important. As India excels in software, we should be able to apply this excellence to robotics as it grows in global significance," he adds.

PCB designing and hardware maintenance are the other skills that are always in demand in the electronics industry.

Where to look

Training institutes not only provide finishing touch to aspiring engineers on their practical skills but also help the industry in improving the productivity of their engineers. EFY Tech Center is one such training institute which offers courses in basic electron-

ics, microcontrollers (ATMEL, PIC and Renesas), robotics, VLSI and PCB designing. It has alliances with leading firms like Agilent and Microchip Technology. Besides conducting regular training at EFY Centers in New Delhi, Bengaluru and Hyderabad, it also undertakes training at many educational institutes.

"Students come out of the colleges with a low exposure to electronic hardware, selection of components and use of test equipment. Project work, which is very important, is either bought or copied. Realising the need for practical training in specific fields that are important to the industry, we offer hands-on training in certain subjects. Learning is through carefully designed presentations and lab exercise. The attempt is to fill the gap by covering what has been missed in the classroom," informs Col. (Retd) N.C. Pande, joint director (training), EFY Tech Center.

"Today, many students who do not get quality education in the colleges are depending on vocational training imparted by training institutes. Such trainings can help in strengthening a student's resume from a job-search perspective. But training institutions cannot lay the foundation, they can only build upwards," states Dr Ravikumar.

According to Kunwer Sachdev, managing director, Su-Kam Power Systems, "Institutes should develop infrastructure in terms of laboratories, equipment and devices to provide students with hands-on experience through experiments in various areas of engineering as per the course curriculum. The interface between the institute and the industry should be very strong; customised programmes may be designed to fulfil the need of the industries."

ISM, a Bengaluru-based training institute, provides training in embedded systems, telecommunications and telecom skills. It claims to impart knowledge and training in areas that are in maximum demand by the indus-

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try and are usually the weakest link in the repertoire of a student.

Sofcon, a Noida-based industrial automation training institute, offers hands-on training/courses in PLC,